

Weather Wise AI

Phase 5: Project Development

Development is done in iterative sprints, building the core (MVP) first and adding advanced features afterwards.

Development Steps

- Environment setup: Git repository, virtual environment, project folder structure.
- Data pipeline: collect API and historical data, clean missing values, engineer features (hour, season, pressure trend, humidity).
- Model building: train a baseline (e.g. Random Forest / Gradient Boosting or LSTM), evaluate with MAE and RMSE, save the best model.
- Backend: build endpoints such as /weather, /forecast, /chat, /alerts; add caching and error handling.
- AI assistant: design prompts that pass forecast data to the LLM and return concise advice.
- Frontend: build the dashboard, charts and chat UI; connect to the APIs.
- Alerts: scheduled job that checks conditions and notifies users.

Sample API Endpoints

Endpoint	Method	Purpose
/api/weather?city=	GET	Current weather and AQI
/api/forecast?city=	GET	AI-refined hourly and daily forecast
/api/chat	POST	Ask the weather assistant a question
/api/alerts	GET/POST	View or create alert preferences

Best Practices

- Commit small and often; use feature branches.
- Store API keys in environment variables, never in code.
- Write modular code with comments and docstrings.